PYTHON PROGRAMMING

Introduction, Installation & First Programs

Learn Python fundamentals, configure your editor, and write and run your first programs.

STUDENT PRESENTATION • WINDOWS + VS CODE

Learning objectives

By the end of this module, you will be able to:

1. Understand Python’s purpose, history and features
2. Install Python and verify the installation
3. Configure VS Code and select a Python interpreter
4. Create, save, run and modify a .py file
5. Recognise basic errors and troubleshoot common issues

1. What is Python?

Python is a high-level, general-purpose programming language.

It lets developers express instructions in readable code and build many kinds of software.

Python is the language; VS Code is where you write and manage your code.

Readable

Syntax is designed to be clear.

General-purpose

Useful for many problem types.

Python files

Source files commonly end with .py.

Python: a brief history and its uses

Created by

Guido van Rossum

First released

1991

Current practice

Use Python 3 for new projects.

Common applications

Web Development

Websites and APIs

Automation

Repetitive tasks

Data

Analysis and visualisation

AI / ML

Models and intelligent apps

Testing

Automated tests

Science

Calculations and simulations

Key features of Python

Readable syntax

Indentation defines code blocks.

High-level

Hides many low-level details.

Dynamic typing

Names can refer to values of different types.

Object-oriented

Supports classes, objects and inheritance.

Standard library

Built-in modules for common tasks.

Cross-platform

Available on Windows, macOS and Linux.

Large ecosystem

Many libraries support specialised tasks.

Open source

Open-source language with a broad community.

Python, interpreter and VS Code

Python language

The rules and syntax used to write instructions.

Python interpreter

Runs the program.

VS Code

Editor for writing and managing code. The Python extension adds development features.

WRITE CODE → RUN WITH PYTHON → SEE OUTPUT

CPython typically compiles source code to bytecode before execution. This happens behind the scenes.

2. Install Python

Install Python before configuring the editor.

1. Open https://www.python.org/downloads/
2. Download a suitable stable Python 3 release for Windows.
3. Run the installer and follow the prompts.
4. If offered, enable “Add Python to PATH” where appropriate.
5. Complete installation, then open a new terminal.

Python and VS Code are separate installations.

Verify the Python installation

In Command Prompt or a terminal, run:

```
python --version
```

If that command is not recognised, try:

```
py --version
```

Expected result

A Python 3 version number appears. The exact version depends on installation.

What is PATH?

An operating-system setting that helps the terminal find commands such as python.

If neither command works, check the installation, PATH and Windows app execution aliases with the trainer.

3. Install VS Code and the Python extension

1. Download VS Code from https://code.visualstudio.com/
2. Install and open VS Code.
3. Open the Extensions panel.
4. Search for “Python” and install Microsoft’s Python extension.
5. The extension adds Python features; it does not install Python itself.

> **Remember**
>
> Python runs the code. VS Code helps you write, organise and run it.

Set up your VS Code workspace

1. Create a folder

Create a folder named PythonTraining.

2. Open the folder

File → Open Folder → PythonTraining.

3. Select Python

Ctrl+Shift+P → Python: Select Interpreter.

Key VS Code areas

Explorer	Files and folders
Editor	Write and change code
Terminal	Run commands and view output
Run Python File	Run the current .py file

Interpreter selection

Choose the Python version approved for the course. If several versions appear, ask the trainer.

4. Write your first Python program

Create a file named hello.py in the PythonTraining folder.

```
print("Hello, World!")
```

Run it using either method:

Option A

Click “Run Python File” in VS Code.

Option B

In the terminal: python hello.py

Expected output

```
Hello, World!
```

Save first: Ctrl+S

Print statements and simple variables

print()

Displays text or values in the terminal.

Multiple lines

```
print("Welcome to Python")
        print("I can run code")
                  OUTPUT
             Welcome to Python

I can run code
```

A variable example

```
student_name = "Alex"
    print("Student:", student_name)
                  OUTPUT

Student: Alex
```

A variable is a name that refers to a value. Variables and data types will be covered in more detail later.

The programming cycle: edit → save → run



Write or change code

Save with Ctrl+S

Run the .py file

Read the output

Fix issues and repeat

Practice

Change the text or student name, save the file, run it again, and confirm the output changes.

Basic errors: read the message, then investigate

Syntax error

Code breaks Python’s writing rules, e.g. a missing quotation mark.

Name error

The program refers to a name Python cannot find.

File / path issue

The filename or terminal folder may be incorrect.

Mistake: missing quotation mark

```
print("Hello, World!)
```

Corrected version

```
print("Hello, World!")
```

First checks: save the file, check spelling and punctuation, confirm filename and folder, then read the full error message.

Common problems and first checks

Problem	First checks
“python” not recognised	Try py --version; check installation and PATH; reopen the terminal.
Output did not change	Save the file (Ctrl+S) and run it again.
File not found	Check the filename and current terminal folder.
Wrong Python version	Use Python: Select Interpreter and choose the course version.
Red underline in editor	Check spelling, quotes, brackets and indentation; some warnings do not stop execution.

Ask the trainer if you are unsure about changing system settings.

Hands-on: complete your setup

Complete each step independently, then demonstrate your result.

- [ ] Verify Python with python --version or py --version.
- [ ] Install VS Code and the Microsoft Python extension.
- [ ] Open PythonTraining and select the interpreter.
- [ ] Create hello.py and run “Hello, World!”.
- [ ] Create about_me.py using print() and a variable.
- [ ] Change code, save it, rerun it and show the updated output.

Module review

Installation

Display the installed Python version.

Configuration

Select the intended interpreter.

File creation

Create a correctly named .py file.

Execution

Run the file and find its output.

Modification

Change code, save and rerun.

Troubleshooting

Check the file, folder and error message.

Next topic: variables, data types and operators.

Official references

Python downloads	https://www.python.org/downloads/
Python documentation	https://docs.python.org/3/
Visual Studio Code	https://code.visualstudio.com/
Python in VS Code	https://code.visualstudio.com/docs/languages/python

Module goal

Install Python, configure VS Code, and write and execute simple Python programs.